Mapping Circular economy and the role of Design in Portugal

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After a long period of maturation, the European commission approved an action plan for Circular Economy in 2015 (EC, 2015). The plan was received and approved by the EU countries, who followed up with national plans. Once the policies have been set, it is also important to recognise the concrete changes they are triggering in the socio-economic context, and, from the perspective of a designer, to understand the possible role of design in such change. This suggests two research questions that can be analysed in relation to CE initiatives in Portugal. The Portuguese action plan sets the actions for CE at the macro, meso and micro level, thus including actions that concern broad strategic and cultural changes (at the macro level), identify key sectors (at the meso level) and local actions (at the micro level). The authors of this paper analysed the circular economy initiatives in Portugal, that have been recorded in a database issued by the Circular Economy Club. The shift from a linear to a circular economy culture and practice can be read through a classification of different initiatives (Alaerts et al., 2019), which helps recognising and qualifying the dimension of change. The analytical framework proposed in this paper also considers a Multi-Level Perspective (Geels and Schot, 2007), which allows to include some initiatives that could not be framed in Alaerts' classification. The change generated by the new policy extends designers' role to new functions: while the potential for circularity in product design have been largely assimilated in designers' activity, the new rules and the panorama of the ongoing initiatives suggest a possible critical role for designers in infrastructuring change. The inclusion of design in policy actions and their role in translating policy actions into concrete initiatives at the local level is something relatively new, that opens new perspectives and new challenges.

Back to the roots

At the beginning of the last decade, the Ellen McArthur foundation (MacArthur, 2013) defined the Circular Economy (CE from now on) as a complex systemic approach to sustainable production and consumption; although the CE has become a "hot" topic for discussion in the last decade, it represents the maturation of different streams of research on environmental strategies and sustainability in the last 30 years. In 1992, Robert Frosh (R A Frosch, 1992), for example, proposed that a strategic approach towards a sustainable transformation of industrial ecosystems could be based on an analogy between natural cycles and industrial cycles.

In nature the concept of *waste* does not exist as every material that is discarded by an organism, or a natural process becomes a nutrient for other organisms or natural processes. By proposing the concept of *Industrial Ecology*, Frosh opened a new perspective on industrial systems. The concept was taken up by Ayres and Simonis (Ayres and Simonis, 1994) to highlight the circular metabolic processes that industrial processes could adopt,

imitating nature, to reduce the amount of industrial waste. The concept of metabolism is used to indicate the capability of natural ecosystems to *digest* waste from natural processes, transforming them into resources for new processes. Frosch and Gallopoulos (1989) used this concept for industrial systems and proposed that waste from one process could serve as raw material from another, thus reducing the overall industrial impact on the environment.

The concepts of industrial ecology and *industrial metabolism* were a good theoretical foundation in the debate on *Extended Producer Responsibility* (EPR), which extends the responsibility of companies over their own products beyond the point of sale (Gertzakis et al., 2002). Extended Producer Responsibility was introduced in Germany in 1991, as a waste management measure concerning packaging (Cahill et al., 2011); soon other countries followed the initiative, and other sectors, from automotive to photocopiers, seized the need for a deep rethinking not only in the business strategies, but also in the way products were designed (John Gertsakis et al., 1998).

The forced need to reuse the material discarded from the use of product opened a perspective over the whole life cycle and the concept of industrial metabolism highlighted the systemic nature of the initial idea proposed by Frosh. Material cycles could not be limited to a specific industrial sector, in which the material had been produced, but had rather to be extended to the various sectors that could be included in the production and consumption cycle. The life cycle perspective suggested a shift from a linear approach (*from cradle to grave*) to a circular perspective (*from cradle to cradle*) (Braungart and McDonough, 2008).

From industrial systems to production and consumption loops

This short summary of the evolution of different streams of thought around the roots of circular economy reveals that the theoretical ground for rethinking industrial production systems was quite solid at the end of the last century. However, the implementation of the theoretical formulation shows a level of complexity that is still being debated in several aspects, such as the possibility of decoupling economy and growth, and consequently the capability of capitalism to address CE within its own paradigmatic structure (Melles, 2021), or the implementation of CE in urban contexts (Campbell-Johnston et al., 2019). The analysis of products' life reveals that the conditions for circularity were different from product to product: some products could be easily reused or repaired, others could be remanufactured (Giarini and Stahel, 1986; Stahel, 1994), thus creating the conditions for product life extension. Others (like cars, white appliances) could be replaced by new products, given the higher efficiency of the new models in respect to the old ones (Heiskanen, 1996; Stahel, 1994).

The Ellen McArthur foundation's butterfly diagram (MacArthur, 2013) includes several strategies to manage the stock of resources, including natural resources and those resources that have already been embedded in the production and consumption system (Figure 2). The strategies on the right side of the diagram are focusing on products' material components in the perspective of recirculating resources. In this scheme, we could read different levels of intervention, based on the quality of knowledge required to feed the cycle: the larger cycles imply an intervention on the *micro* level, i.e. on materials. Sometimes this intervention concerns the molecular manipulation of material, as for polymeric materials, or mechanical treatments, as for organic materials or simply the destruction of molecular links, as for energy recovery. This intervention is usually reducing the value and quality of materials and requires advanced and specialized, technical knowledge, beside the specific knowledge needed for the activation of material flows and the adaptation of business models. Progressing towards the

narrower cycle, strategies are referring to larger systemic change, which relies on business or socio-cultural knowledge. If we consider the technical knowledge embedded in products (from molecular to mechanical knowledge), the strategies related to the narrower cycles in Figure 1 are those that recover most of the knowledge embedded in products, whereas the larger cycles refer to strategies that imply a higher use of resources (to transport, disassemble and recycle products) and the larger waste of technical knowledge (downcycling): indeed material recycling or incineration imply that all the knowledge embedded in a product is destroyed and only the minimal characteristics of the material or the pure energy is recovered.

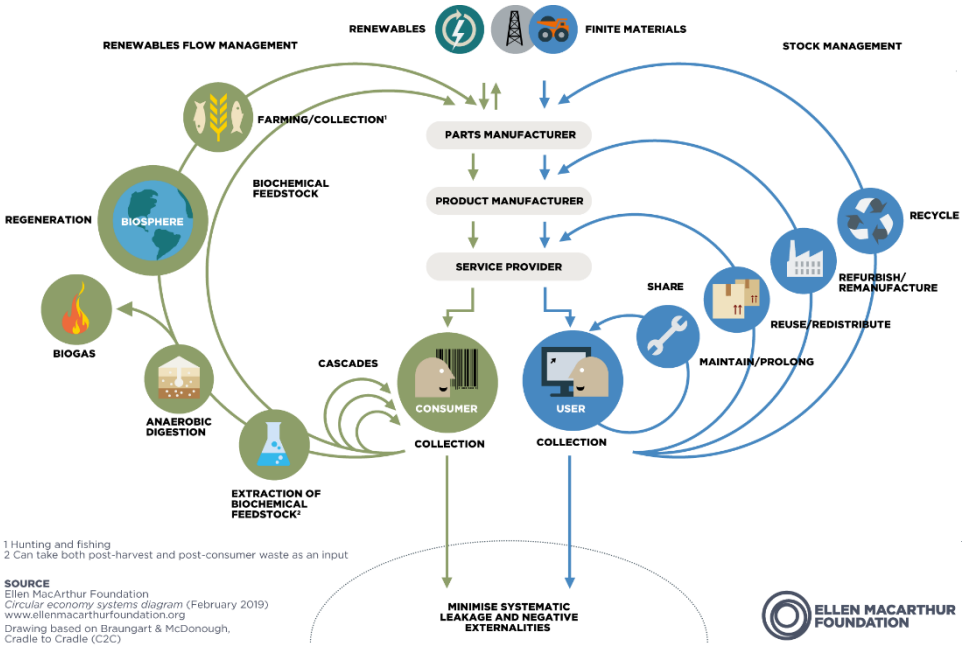


Figure 1. The circular economy system diagram (source (McArthur Foundation)

While the circular use of resources can be easily monitored in larger circles, the narrower circles require a strategic approach that cannot be exemplified through the logic of flows, including strategies to reduce energy use, increase usage intensity and possibly consider different functions to address specific needs.

Alaerts et al (Alaerts et al., 2019) classify circularity strategies from linear to circular economy, taking into consideration the complexity of the initiatives, which require a progression from technological innovation to socio-technical change. By doing this, they consider strategies for a smarter product use and manufacture, which include strategies for addressing users' needs through a range of different functions or to increase the usage intensity or product efficiency.

Table 1 Circularity strategies, adapted from (Alaerts et al., 2019)

Circular economy	**Level A** Smarter product use and manufacture	Refuse	Make the product redundant by shifting function	Socio-technical change
		Rethink	Increasing usage intensity	
		Reduce	Increasing product efficiency by using fewer resources	
	Level B Extend the lifespan of	Re-use	Reusing a discarded product that is still in good condition	

Linear economy		Repair	Repair or maintain defective product	Technological innovation
		Refurbish	Restore and old product bringing it up to date	
		Remanufacture	Re-use part of a discarded product in a new product with the same function	
		Repurpose	Use a discarded product or its parts for a new function	
	Level C Useful application of materials	Recycle	Process materials to obtain new material	
		Recover	Energy recovery	

The different strategies proposed in this case are focused on product use and are implicitly indicating a logical progression from linear processes, mainly connected with technological innovation initiatives, to socio-technical change (see Table 1). This progression excludes the purely linear processes – those in which resources are extracted from nature, used and discarded in nature – but starts from the simple recovery of materials (level C), which corresponds to the larger circles of the butterfly diagram in Figure 1, which focus on materials and energy recovery. The strategies increase their complexity when they imply services, or institutional infrastructure or socio-cultural changes (level B and A) that involve a larger part of the socio-technical system.

Some authors highlighted the systemic aspects related to the strategies for product life extension. In particular Stahel (Stahel, 1997) proposed a *service (cycle) economy*, in which the closure of material loops was proposed in relation to the closure of *liability loops*, i.e. the creation of services to distribute the liabilities/responsibilities in a society working on closed material and energy loops. The service analysed in Stahel's exploration included infrastructure for sharing the utilization of goods or to recycle/remanufacture goods as well as services that referred to systemic sociocultural solutions, such as platforms (the term platform was not used by Stahel, as it was introduced a few years later) to sell results, utilization of goods, in a perspective of a *functional economy* (Giarini and Stahel, 1986) or *Performance Economy* (Stahel, 2010).

The principles of Extended Producer Responsibility and the idea of an extension of the concept of industrial metabolism to larger human systems (such as cities or regions) are so translated into the concept of circularity and mapped onto various concrete aspects of existing socio-technical systems, from product-related strategies aimed at facilitating product disassembly and material recycling (Gertzakis et al., 2002) to product-life extension, which often implies the revision of the principle of product ownership (Stahel, 1997). This debate was extended to a broader discussion between those that call for a total redesign of the whole socio-technical system, based on capitalist principles, and those that believe in the capability of the same system to address CE with a number of adjustments (Melles, 2021).

A multilevel perspective on sustainability transition

The perspective shift from products to their utilization is part of the debate on the nature of change proposed as the basis for circular economy: the initial perspective focused on products suggests strategies to extend product life (through repair or modular configuration) or the life of materials (through product disassembly and recycling). This perspective was mostly geared toward localized changes in product configurations and design. The idea of extending the producer responsibility to the whole product lifecycle, though, implied that no substantial change was possible without the support of an infrastructural system.

In this sense, the systemic analysis of the change required by a perspective of circular use of products can be framed through multi-level sustainability transition theory proposed by Geels and Schot (Geels and Schot, 2007): the perception of a risk connected to the extension of product responsibilities led companies to generate different product-related strategies, but such strategies were fated to stay in a niche, they could only emerge and be extended to their entire industrial sector when *regime conditions* (e.g. policy-driven initiatives, that activate specific services, infrastructures, legal configurations) could be created, which could facilitate the operation of maintenance, repair and recirculation of products, components and materials.(Melles, 2021)

The emergence of such regime conditions, in turn, was only possible once a socio-cultural change forced a substantial shift in the focus of public debate, legislation and institutional structures. The definition of such shift implies cultural changes at different levels. Some authors, such as Stahel (Stahel, 1997), proposed a shift from the intrinsic value of a product to the value of its utilisation. The implications of such shift are cultural, before being technical, as they concern not only single products but social practices and lifestyles, ways of thinking and perceiving reality i.e. what Geels and Schot define as a *landscape change*. The outcome of the shift proposed by Stahel would therefore concern both the socio-technical regime and the socio-technical landscape in Geels and Shot's multilevel perspective (Geels and Schot, 2007).

Mapping the CE landscape

The evolution from the earliest concepts related to sustainability to the concept of Circular Economy is visible in different dimensions: one dimension concerned the expansion from the initial focus on critical sectors (packaging, automotive) to all the areas related to industrial production (Howard et al., 2022; Emf, 2021); another dimension concerned the expansion from the initial focus on extended produced responsibility to a redesign approach (Ceschin and Gaziulusoy, 2019, Simeone et al., 2019), which could facilitate maintenance, repair, remanufacturing and material recycling; a third dimension concerned the extension of the focus from products to their use or functionality (Stahel, 2010), which was focusing on infrastructures (services) and socio-cultural changes that would make circular economy possible.

The concept of CE was probably the catalyst that inspired a number of initiatives in the last decades, which populated a panorama of niche innovations (from products to services) infrastructural changes (local, national and international legislation) and socio-cultural changes. Among policy innovations, the European Green Deal proposed a large socio-cultural change that was expressed through a roadmap and, more specifically, the European Circular Economy Action Plan (EC, 2020) was setting the regime conditions for the CE in different sectors. At the same time, the Green Deal itself is the sign of a large landscape transition towards an institutional culture in which the question of sustainability becomes crucial to any development of society, economy and technology. The result of the change occurred in the last ten years is in the wide range of initiatives undertaken in every sector, with the aim of exploring the multiple dimensions that the concept of CE implies.

The various dimensions described so far propose a good basis for mapping CE initiatives in a specific local, regional, national or international context. The authors of this paper, however, tried to go one step further and to map a specific landscape, i.e., a specific national context, to address two research questions:

- How can we recognize and qualify the different dimensions of changes towards CE in a real context? This question implies an observation of how the change related to CE is actually happening in this context. This observation, in turn, is particularly relevant to inspire future initiatives and could also address a second important research question.
- How can a change towards CE economy be designed? This question is particularly relevant from two perspectives. From a perspective focusing on socio-technical systems, it is important to see to what extent the systemic change related to circular economy can be addressed, i.e. can be designed. This means taking into account how favorable developments can be included to enhance the horizontal scalability of relevant niches or the vertical scalability (i.e. the institutionalization) of some preferrable changes. From a designer's perspective, the question should introduce a reflection on the role that design can have in such a change and on the design competences that can support it.

The focus on Portugal as the main case

The national context analyzed in this paper is Portugal. The choice has been suggested by the location of one of the authors and, therefore, by the possibility to access relevant empirical material. It is worth noticing that the information on this context is by no means complete, as cases and initiatives on CE are continuously developing. However, the choice of a specific set of data related to this context makes it possible to refine an approach for reading and interpreting change in the light of CE principles and perspectives.

Since 2014, the EU has promoted CE strategies through various projects and initiatives, the most recent and robust of which being the 2018 EU Circular Economy Package (Bourguignon, 2018). The EU also promoted an action plan for circular economy, aimed at ensuring a regulatory framework for the mobilization of economic operators and society at large in promoting initiatives towards waste treatment strategies oriented towards circularity. The EU action plan focused on actions at EU level, but highlighted the need for long-term involvement at all levels, including states, regions and cities (EC, 2015).
In 2017, Portugal approved the Action Plan for CE, The Portuguese action plan ("Leading the transition", 2017) sets the action at the macro, meso (sectorial) e micro (regional/local) levels. The macro level includes measures at the strategic level, which focus on seven different aspects, such as extended producer responsibility, education, food waste reduction, resource regeneration and research and innovation. The implementation of these initiatives was based on political instruments such as green taxation or voluntary agreements; at the sectoral and regional levels, other incentives were put in place, i.e. funding, that could support the development of solutions e.g. fundo ambiental, fundo para a inovação, tecnologia e economia circular, Portugal 2020 funding program.

The meso level sets the agendas for action, concerning the efficiency of material productivity and public procurement. At this level, the action plan indicates key sectors of the Portuguese economy such as tourism, textile, shoe industry, retail and resource intensive sectors (construction). Finally, the micro level focuses on actions at the local level, setting expectations for initiatives from companies, local administrations or other local stakeholders (ETC/WMGE, 2019); at this level the agendas focus on developing circular cities, organizations/companies and promote industrial symbiosis. The action plan also triggered reflections and studies on *industrial symbiosis* as a strategic approach to make sure that waste from a production/consumption chain could be reintroduced into another productive chain (BCSD, 2018).

In early November 2022, the evaluation of the implementation of the plan and the public consultancy of research results was undertaken to support the development of the second Action Plan for CE in the country (LNEG, 2022). Seven dimensions for action were proposed, identifying driving forces and barriers to further infuse CE in culture, investment, technology and infrastructure, policy and regulation, stakeholders network as well as institutional and organizational dimensions. Despite being identified as a leading country in terms of circularity initiatives (Winans et al., 2022), major barriers regarding design infusion were identified, especially regarding the "lack of infrastructures that enable circularity"; "lack of investment that still exists in ecodesign", "products that are not prepared for circularity" (LNEG, 2022). As mentioned by the founder of the Circular Economy Portugal Association, and consultancy advisor for the government, in conversation with one of the co-authors: "circular design needs to be more emphasized in this second action plan" (mentioned during a project meeting with the co-author).

Circular Economy initiatives in Portugal

To help us identify the initiatives, this paper includes the analysis of a series of CE initiatives developed in Portugal based on the information collected from a large database issued by the Circular Economy Club (CEC) – an international non-profit network, including professionals and organizations with over 280 CEC local clubs in 140 countries. Portugal has been active since 2020 and has currently 7 active clubs scattered throughout the country namely in Porto, Braga, Felgueiras (northern region), Lisbon and Almada (center) and Faro (southern region). The study of the Circular Economy Club refers to a mapping week in February 2018, which engaged 2100 participants from 40 countries through 65 workshops. The workshops developed a database of 3.000 initiatives related to CE, but the database has been further developed after the mapping week. Operating from this database, the authors identified 98 initiatives related to Portugal.

The database included different aspects collected by the participants of the workshops of the Circular Economy Clubs. One of them consisted in the identification of (1) six CE strategies namely, design, resources, business models (BM), product life extension (PLE), waste as resource and other strategies and (2) six main sectors in which innovation appeared such as cities (including buildings, infrastructure, mobility), consumer products and electronics, fashion and textiles, food and beverage, manufacturing, and other sectors (including financial services, healthcare, education, and tourism). Finally (3) the database identified the type of organization leading the CE initiatives: Multinational Corporate, National Corporate, SME, Startup, Private sector, Education, Non-profit, Government, Support (investment, consulting, media, etc), and other.

Preliminary results of existing data and categories

The analysis of preliminary results show that Portugal appears at the 12th place in the ranking (3,24%) of the CE initiatives mapped by the CEC around the world, with 50 participant countries. The country is preceded by the UK (with 13% of representation), Sweden, Germany, Canada, South Africa (between 5-10% representation); and Taiwan, Austria, Denmark, Spain, USA and Chile (between 3,5% to 5% representation).

In comparison to the rest of the world, Portugal has identified more initiatives in specific key sectors of its economy, namely in Fashion and Food/Beverages (with approximately +5% in each organizational sector) (Figure 2). Organizational sectors with less representation are Manufacturing (-6%) and Cities (-4%).

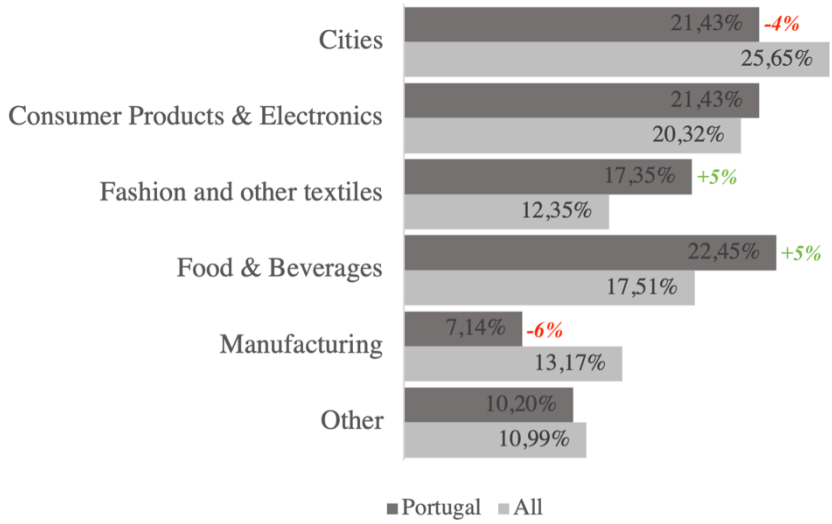


Figure 2. CE initiatives by sector

Additionally, when analyzing the type of organizations involved in the CE initiatives, a major difference of representativeness of the private sector (-19%) and SMEs (-16%) was identified, when compared to other countries. In contrast, a higher number of Non-profit (+8%), National Corporate (+9% approximately), Startups (+11%) CE initiatives in the Food/Beverage (for Non-profit and National corporate) and Fashion sectors (for the Startups) were identified in the database (Figure 3).

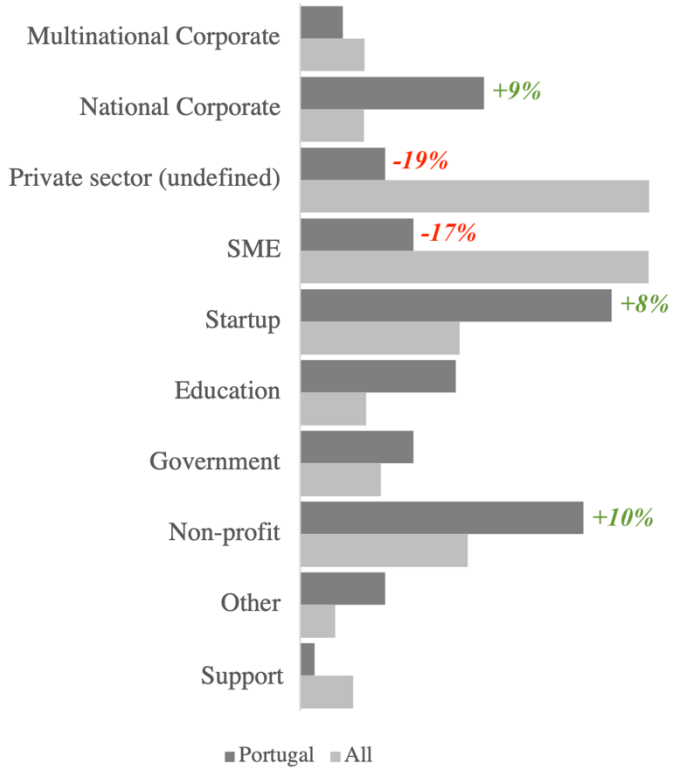


Figure 3. CE initiatives by actors

Finally, when compared to other countries, the primary CE strategies in Portugal are more focused on designing new business models (Figure 4). This type of strategy is mostly used in the consumer products/electronics sector.

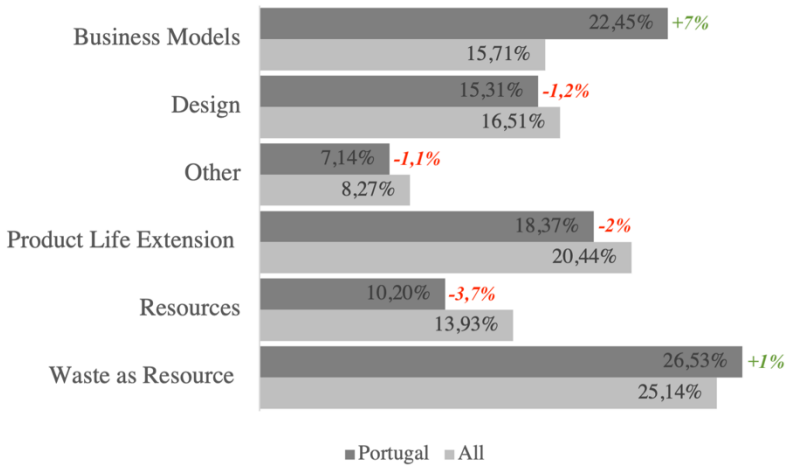


Figure 4. CE initiatives by CE primary strategy

When highlighting the analysis of the CE initiatives collected in Portugal specifically, the main organizational sectors involved in such initiatives are food/beverages (22%). A large part (21%) of the initiatives were also concerning urban components (Cities - buildings, infrastructure, mobility, logistic, resource management) and Consumer Products/Electronics (21%). The Fashion sector is also included with 17% of initiatives and manufacturing has the lowest representation (7% of cases).

Second, up to one third of the organizations/entities leading CE activities are Startups (~23%) and Non-profit organizations (~20%). This data can indicate that the institutionalization of CE may be still at an early stage in the country (Salamzadeh and Kesim, 2015), as startups are newly born companies which often struggle to consolidate their business in the mainstream market. The ideas and prototypes are still at the testing stages, often construed with high uncertainty of success; the high participation of non-profit organizations in CE initiatives can also indicate that CE may yet not be profitable and still be limited to small niches, with limited potential for scaling out (replicating) (Moore and Riddell, 2015) such initiatives in other contexts. Some initiatives do rely partially on voluntary citizens' movement and participation, finding alternative ways of doing/living, which are complementary (but sometimes not compatible) with market growth (e.g. permaculture).

Third, the most recurrent CE strategy used in Portugal, according to this database, concerns the reuse of Waste as a Resource (~27% of the initiatives concern this strategy); another substantial quote (22%) of initiatives concerns the development of a New Business Models. Product Life Extension and Product Redesign were also recurring with a lower frequency (18% and 15%, respectively). When crossing the different categories as seen in Figure 5, Waste as Resource CE strategy is embedded in initiatives identified throughout the different sectors. Other strategies, however, seem to be more secluded to specific sectors. For example, “Design” is used mostly in the fashion and manufacturing sectors. “PLE” is mostly used in Consumer Products and food/beverages sectors. “Business Models” strategy is identified in four sectors – but is more intensively used in the Consumer Products sector.

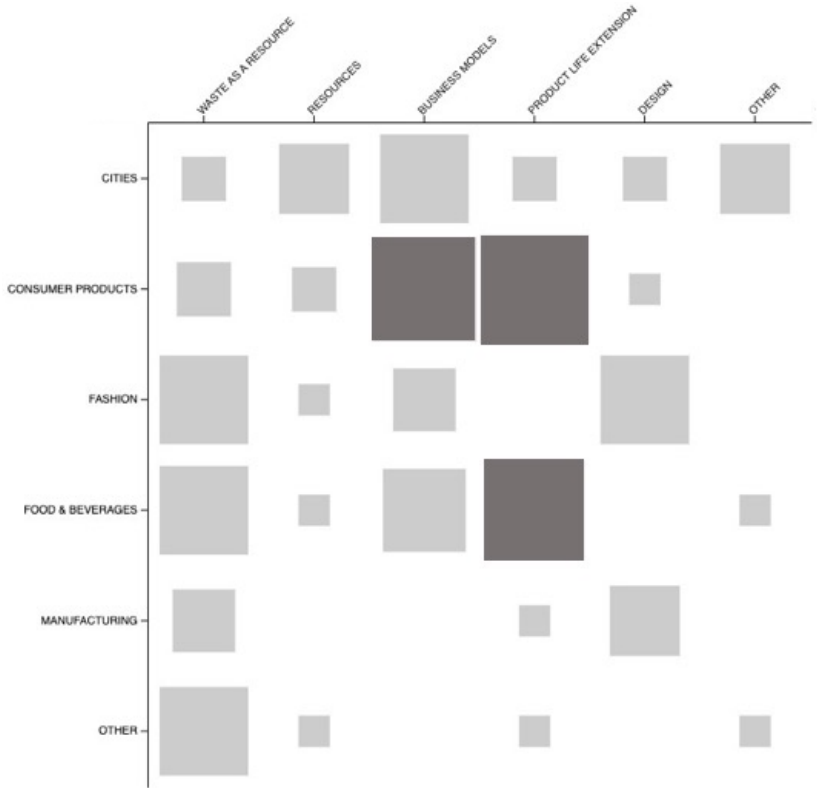


Figure 5. CE primary strategy and organizational sectors

In summary, the report related to the database includes some categorizations that are useful to read the change CE is defining in societies. The interpretation of the model, however, does not provide useful insights to understand the transitional value of CE initiatives, i.e. to understand how such initiatives are triggering change at the micro level (e.g. by promoting a single technical initiative), or at the meso level (e.g. by generating a system based on a series of services or social interactions between different stakeholders) or at the macro level (i.e. at the level of major socio-technical changes).

Some considerations about the database

While the database provides a good *logical geography* to navigate and understand the development of CE initiatives in Portugal, it does not provide a key to understand the potential impact of such initiatives. On its own, this database does not help understand the evolution of CE in Portugal

A first consideration should concern the scale of innovation, which in turn is connected to the complexity of the initiative.

Because of its comprehensiveness, the strategy framework proposed by Alaerts et al. (Alaerts et al., 2019) (see Table 1) can be used as a first filter to interpret the database of initiatives in Portugal. The authors of this paper tried to map the criteria proposed on the framework on the initiatives in the database. While the mapping is purely based on the authors' interpretation of the database, it gives a first glance of the distribution of the initiatives along different levels of complexity. It is worth mentioning, though, that some initiatives could not be included in the framework: the database indeed, reported initiatives related to the creation of a) educational initiatives, aimed at creating capacities (i.e. diffuse knowledge among new generations) and b) policy or government actions. For this reason, two new categories have been added to Alaerts et al.' categorization: education and policies and government actions.

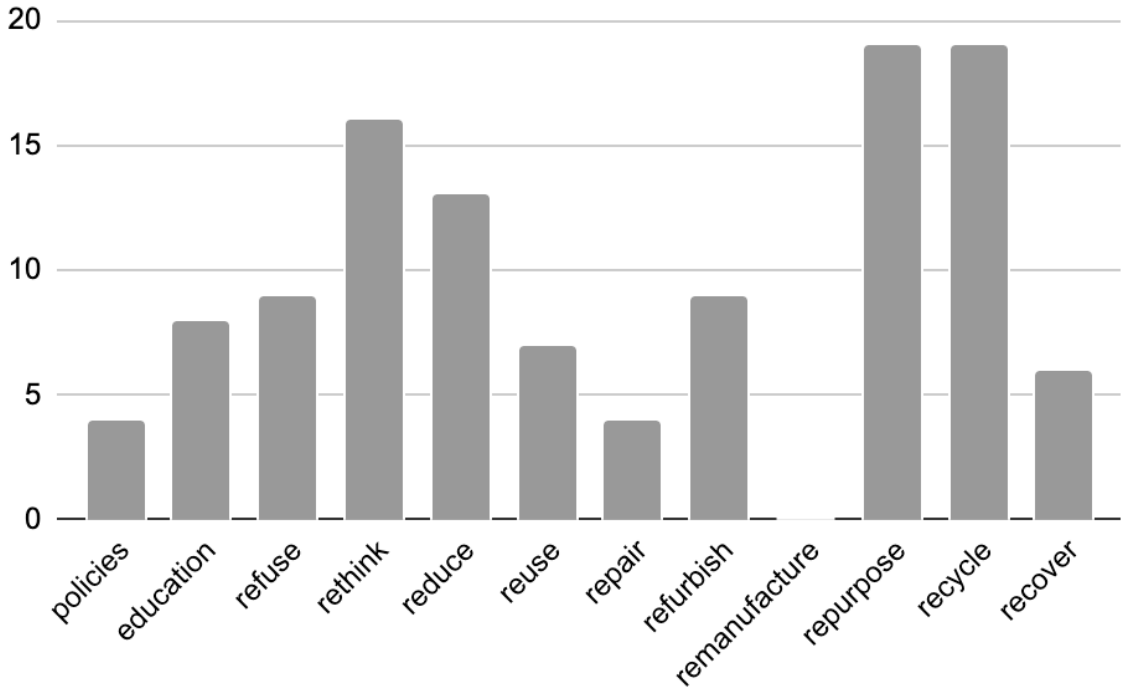


Figure 6 CE Strategies in Portugal

Figure 6 provides a visual overview of the strategies, as they have been identified in the database. The figure shows how the most recurrent strategies concern the activities of recycling and repurposing, with a lower incidence of initiatives aimed at recovering energy from discarded materials, which is destroying most of the properties, knowledge and characteristics of products. Interestingly, many initiatives refer to systemic aspects, i.e. to the higher part of Table 1. Those are initiatives to exchange second-hand products, sometimes creating marketplaces, or to use less resources or re-focusing on functions, thus shifting to different production and consumption configurations.

Beyond the data

To address the first research question proposed in this paper (*how can we recognize and qualify the different dimensions of changes in a real context?*) a further step is needed: it is important to *enter* into the data and investigate into the nature of the initiatives. For this purpose, the authors collected further information about the initiatives, using the links and documentation reported in the database, and interrogated the database according to different perspectives, using additional labels that open to different categorizations.

One of the most meaningful characterizations concerns the focus of the initiative: the landscape observed through the database included initiatives of very different nature, from intervention on buildings to education, from redesign of a product to the definition of marketplaces for second-hand products and to the definition of policies to support circular economy. The authors used different labels, possibly attributing more than one label to the same initiatives.

Table 5 Main focus of the initiatives

Focus	**Description**
Building	Initiatives that concern the building sector
Education	Initiatives for schools or training actions or actions that increase the awareness of possible circular flows in a community
Infrastructuring	Initiatives that generate opportunities for a smarter use of products and services or that support the extension of products lifespan. Within infrastructuring, further subcategories have been identified: - Platforms: creation of opportunities for material reuse, swap marketplaces, multiple use of products (sharing) - Events: organization of fairs, periodic markets, demonstration events that increase citizens' awareness of the opportunities for circular actions - Services: modification of existing services to optimize the use of resources (e.g. reducing packaging)
Policies or government actions	- Policies: policy actions aimed at creating roadmaps or subsidizing circular economy initiatives
Product	Initiatives that focus on products, to increase their lifespan (repairing, reusing, remanufacturing) or to optimize the use of materials.
Resources	Initiatives for the reutilization of material of for the optimization of resources (e.g. waste into energy or water)

The foci defined in Table 5 could be further grouped:

- A group of initiatives that focuses on the physical or molecular manipulation of materials (e.g. recycling) or resources (e.g. water saving). Those initiatives are mostly related to the level C in the classification proposed by Alaerts et al. (2019). This level includes innovation related to buildings, products and resource management. Most of those initiatives rely on changes based on technologies that make it possible to reuse materials or resources (from cigarette butts to water or codfish spines). The database does not provide clear clues on the reason why those initiatives are more diffused than others at level A or B; a hypothesis could be that those initiatives can be better controlled by one or few actors (those who handle the technology), with reduced involvement of other actors (such as household, communities or schools), that usually have a role in the collection of discarded products (such as electronics or food). Since they do not require major behavioral change, those initiatives may be easy to establish and consolidate.
- A group of initiatives that works on the creation of exchange platforms, marketplaces, special events or services; these are initiatives that create the conditions for re-use, refurbish, repurpose or repair (e.g. material repositories, childrens' toy sharing) and for rethinking the way products are used (e.g. car sharing). These initiatives are related to the levels A and B in Alaerts' classification. On the other hand, these initiatives could be defined as *infrastructuring* initiatives, as they generate *regime conditions* for activating flows for exchange or recirculation of products, components or materials.

- A third group of initiatives concerns initiatives aimed at generating capacities at the systemic level, by proposing specific policies or by activating education initiatives that tend to trigger cultural change.

It is worth noticing that this grouping provides some clues to compare the observed innovation landscape to the multilevel structure proposed by Geels and Schot (Geels and Schot, 2007):

1. rather than proposing any systemic action, the initiatives in the first group focus on specific materials or products or resources, thus providing solutions that concern a specific niche, with its problematic implications. Those initiatives often imply the use of specialized technical knowledge (from architects to chemical engineers), and they represent a good opportunity to advance knowledge on specific technical problems. For example, the "redbridge school" project uses wood and other renewable or low-impact materials to build the structures and facades of the school. In the fashion sector, a well-known shoe brand "As Portuguesas" uses residues of the cork from the stopper industry as the main raw material to create the shoe soles – thus using existing materials to generate new value via product design. Finally, looking at resources, other examples can be identified, to recycle or recover material, namely, the initiative developed by EPAL in Lisbon, that uses treats wastewater to use in public gardens maintenance, or cleaning public facilities and goods such as metro stations or buses.

2. the infrastructuring initiatives concern the creation of swap markets or (sometimes local) marketplaces or platforms, which activate new practices, thus new socio-technical routines, that can change the *regime* conditions in the Portuguese context. It is worth noticing, however, that a regime change, as described by Geels and Schot (Geels and Schot, 2007) includes changes in the whole technical and institutional infrastructure of a context, which cannot be observed from the initiatives in this database, possibly because such change cannot be observed from any single initiative but is rather a systemic emergence. This group of initiatives mostly involves actors with different skills, including design, communication, or social skills. The infrastructuring initiatives in the dataset at hand appear to have a wider spectrum of intervention types – from creating platforms to events, services and policies, at the local and national levels. For example, initiatives related with "events" are mostly focused on organizing markets for reselling or exchanging second-hands items; and/or service events to repair appliances/objects. Services include initiatives related with "smarter product use and manufacture", supporting citizens to extend the products' life, namely for system needs related with mobility (car sharing, bike sharing), as well as with education (centers for collecting and sharing schoolbooks).
   The generation of platforms are also a strong mechanism to support infrastructuring towards CE. Platforms can be defined as a "set of actors, resources, business and operational rules and setting that converge for value creation" (F. Breidbach et al., 2014)
   Although platforms are often digitally enabled (e.g. Booking Drive, an online shop to rent all kind of equipment), the database also provides a set of examples of alternative mechanisms to adjust and develop the conditions for smarter product/service use and needs fulfillment, for example, by creating alternative channels for zero waste of food and clothes (e.g. Fruta Feia that works with wasted/ugly fruits/veggies, which do not fit market requirement for retails, are sold to end customers based on a monthly fee); or chain of shops to sell/buy or exchange second-hand goods/furniture.

3. Finally, the third group of initiatives aims at a broader change in the socio-technical system. If effective, those initiatives will impact on the institutional system and on the cultural landscape of the country, therefore they could be compared to changes in what Geels and Schot defines as the socio-technical landscape. Obviously, the database only offers a perspective on the potential landscape change, rather than a real view on something really happening. Policy initiatives include the definition of roadmaps or action plans for the public support to CE at the municipal as well as the national level. The potential of such an initiative is in the capability to create a seamless support to an otherwise fragmentary landscape of small changes that can be observed through the database. Education is also an important vector for capacity building, that spreads throughout tall the levels of complexity related to CE, including:
    - initiatives to promote CE in schools and public spaces, creating education programs, courses and providing training to change existing practices (level A in Table 1), Examples include "escolas circulares" (circular schools), or "vegetable garden at your door" that provides citizens with garden plots to practice organic agriculture and receive training in farming and composting;
    - educational initiatives that address repair or refurbish (level B in Table 1), such as the RLAB, a knowledge laboratory to learn to recover electric/electronic equipment;
    - educational activities that support the reduction of centralized treatment of organic waste or workshops about composting (level C in Table 1). The long-term impact of these initiatives in consumer behavior and market changes, through time, is however, still a topic that requires further inquiry.

The cases mentioned in this section can be consulted in the file annexed to this chapter. They are also available and updated regularly in the Circular Economy Club open-access repositories (https://www.circulareconomyclub.com/gd-home/cec-global-database/).

Designing circular economy

The considerations proposed in the previous sessions provide clues to address the second research question: whether (and to what extent) circular economy can be designed. The systemic nature of CE implies that several actors and several competences must be involved in the process of change. Technical competences are crucial for the process of change, especially regarding the manipulation of physical and chemical composition of materials and products; however, as seen in the observation of the database, other competences are also crucial, because they generate societal conditions for change, in organizational, social and cultural terms. Many actors are able to contribute to such change, and therefore, according to Simon's (Simon, 1969) definition many actors are *designing* this change according to their own competences. Those that, according to Manzini (Manzini, 2015), could be defined as expert designers, can contribute to the initiatives in the various groups identified in the database with different competences.

The initiatives focusing on products and materials require designers to redesign products, to reuse materials from previous cycles or facilitating product disassembly or modularity. This area is not new for designers, as it was perceived as a critical challenge to approach Extended Producer Responsibility and therefore developed with several contributions for several decades (Ceschin and Gaziulusoy, 2019; Lewis et al., 2001).The initiatives focusing on

infrastructuring are urging designers to find solutions that challenge the existing socio-technical system. The three main areas defined in Table 5 are: platforms, events and services.

Platforms generate opportunities for interaction among different stakeholders: they could consist of online or local swap markets that recirculate used products or materials or even useful knowledge. The level of complexity can be quite different. Some of the initiatives, such as *Fashion* Revolution are relying on existing social networks for the interaction among the users, while they provide useful information in their website. Other initiatives are in fact promoting the exchange of material (e.g. "Repositório de materiais" – Repository for materials), but, because of the nature of the material exchanged (building and construction material) need to focus on specific geographical areas, possibly planning to scale out by nodes, i.e. generating local material exchange networks in different cities. Finally other platform initiatives are based on an accurate design of the interaction between different stakeholders: *Refood*, for instance has created an ecosystem that includes supermarkets, restaurants, food producers, voluntaries and other actors to produce meals for persons in needs and institutions. The design of a platform requires a good understanding of systemic aspects, including the interaction between different stakeholders and their mutual negotiation. As seen in the database, platforms can emerge from social interaction within existing challenges, but they can also have higher level of complexity, in which the interaction, i.e. the way value is exchanged among the various stakeholders, needs to be accurately organized. In respect to this, the effort of product/service design should therefore focus on understanding the complexity of such systemic aspects: designers should focus on generating appropriate frameworks for action, rather than specific solutions (EC, 2020).

Events are temporary opportunities for product exchanges (e.g. flea markets) or for getting knowledge on how to repair or extend products' life. The social nature of such events makes it possible to organize them through the most common social channels, or by using existing networks that have been created around fablabs or other active organizations. They are obviously local events, even though initiatives like bike repair or repair cafés are increasingly replicated in several locations. (e.g. Repair café Lisboa). Although some of those events are the result of a *diffuse* design capability (Manzini, 2015), the way in which those initiatives are replicated or scaled out is an interesting design problem, as they may imply a capability to understand contexts, engage relevant stakeholders and facilitate social aggregation (Morelli, 2014; Concilio et al., 2013; Morelli, 2015).

Because of their interactive nature, platforms and events can be often considered as **services**, i.e. structured solutions that mobilize a number of stakeholders and technologies to address specific needs related to the utilization of products, or create specific value for different beneficiaries. The database however, also includes initiatives consisting in specific solutions that support the reduction in resources or material use (e.g. the suppression of packaging) and or the rental of goods that are not frequently used, such as luggage or baby equipment.

The initiatives aimed at changing the socio-technical landscape to support CE include educational initiatives and policy or government actions. While designers have been involved in different educational initiatives, the participation of design to policy actions or to institutional structures is relatively new and open new perspectives for designers. The initiatives in the database define roadmaps or action plans for the intervention of public institutions into relevant environmental problems. Designers are increasingly involved in this strategic process (Whicher et al., 2018). The definition of missions is becoming parts of the strategic planning at different institutional levels (Mazzucato, 2018; Miedzinski et al., 2019)

and the European Green Deal EC (2019) is a relevant outcome of this approach. The definition of research and innovation programs at the EU level, clearly recognizes the role of design in coordinating and synergising different elements of innovation through visions and consistent scenarios, but also in translating the elements of such visions into concrete projects. While it is not possible to identify this role in any specific initiative in the database, it is possible to figure out the role design can have when translating the wide program of the Portuguese roadmap for the circular economy into initiatives at the local level, promoting operative action and participatory initiatives to support sustainable plans at the municipal level.

Conclusions

CE is clearly calling for a broad systemic view, which links initiatives in niche sectors or on specific products or materials to larger infrastructural changes and to systemic initiatives, which orient the socio-technical landscape towards circular flows. This paper tried to read a "real" context, as described by a large database of initiatives, through the lens of multilevel innovation, as proposed by Geels and Schot; the idea, in using this filter, was to make sense of the data in the database. Geels and Schot's perspective is not completely adequate to interpret the database (or at least the database does not offer enough information to relate the situation to the Geels and Schot' perspective), however it gives the spark to an analysis of a context that would otherwise seem too articulated and complex. It also gives a chance to highlight some initiatives that go beyond the traditional action of designers, following the evolution of designers' professional domain, from products to services and, more recently to systemic and institutional perspectives.

The challenge that CE proposes to designers is to define their action through a full palette of capabilities (Morelli et al., 2020). While in the early phases of the debate on sustainability designers were working around products and their material components, they are now asked to use their capability to activate social processes, to understand and support mechanisms of scalability, to work across different levels of complexity and to translate broad visions about the future of this society into operative projects and organizational structures.

It is important to take into account that the database analysed in this paper is to be considered as a sample, rather than the complete description of the innovation landscape around CE in Portugal. The whole landscape is continuously changing, and new initiatives are being proposed. The database only represents a ground for discussion on a) how a myriad of innovation phenomena at different scales is shaping broad changes; b) how this change can somehow be oriented through a design action and; c) what kind of challenges designers are facing, when entering in the mechanisms of change activated by CE.

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